BIOCHEMISTRY CUMULATIVE EXAM

April 9, 1994

a. (15 points) In following the elution of proteins from a column, the absorbance at 280 nm, A_{280}, is generally measured on collected fractions or during the process of elution to locate protein bands. What structural features or components of proteins are responsible for the absorbance at 280 nm? List at least three in decreasing order of their expected contribution to the absorbance at 280 nm.

b. (10 points) Do all proteins have an absorbance maximum in the UV range of exactly 280 nm? Why or why not?

c. (15 points) Do all proteins have the same extinction coefficient for absorbance at their UV absorption maximum? Explain your answer. How would you go about determining the extinction coefficient for a given protein at this wavelength?

d. (15 points) Although most proteins absorb at 280 nm, other molecules of biological origin can also absorb at this wavelength. Give an example. How might you determine whether a peak collected from a column was a protein or this other material?

e. (15 points) Although A_{280} is often used to monitor protein elution during column runs, the absorbance at 210 or 220 nm is often used to monitor the elution of peptides during HPLC peptide mapping. Why is this wavelength used rather than 280 nm? What is responsible for absorbance in the 210-220 nm range?

f. (15 points) Of the 20 amino acids found in proteins, only one is normally fluorescent. Which one is it? Although the detection limits for fluorescence measurements are lower than for absorbance measurements, fluorescence measurements are seldom used to monitor elution of proteins from columns. Suggest two reasons why?

g. (15 points) Other than the use of A_{280} or fluorescence measurements, describe at least two other methods or measurements which might be used to locate a protein being eluted from a column.